

Lecture 6: Subnetting Example & VLAN

Part 1: Subnetting — Worked Example

Question: Subnet the Class C network 205.13.221.0 to create 6 LANs, each with at least 25 hosts.

1. How many bits to borrow?

- We need 6 subnets. Test each borrow amount using 2^n patterns:
 - Borrow 1 bit → 2 subnets — not enough
 - Borrow 2 bits → 4 subnets — not enough
 - Borrow 3 bits → 8 subnets — enough! (we only need 6)

2. Subnet Mask

- Default network bits (Class C) = 24. Add 3 borrowed bits → 27 network bits.
- Binary mask: 1111 1111.1111 1111 1111.1110 0000

 **Answer: Subnet Mask = 255.255.255.224 (also written as /27)**

3. Maximum hosts per LAN

- Before subnetting: 8 host bits (last octet).
- After subnetting: 5 host bits remain ($8 - 3$ borrowed = 5).
- Usable hosts = $2^5 - 2 = 30$ (subtract network & broadcast address).

 **Answer: Maximum 30 usable hosts per LAN**

4. The 8 Subnets Created

Each subnet is a block of 32 addresses ($256 \div 8 = 32$).

#	Subnet Address	First Usable Host	Last Usable Host	Broadcast Address
1	205.13.221.0	205.13.221.1	205.13.221.30	205.13.221.31
2	205.13.221.32	205.13.221.33	205.13.221.62	205.13.221.63
3	205.13.221.64	205.13.221.65	205.13.221.94	205.13.221.95
4	205.13.221.96	205.13.221.97	205.13.221.126	205.13.221.127
5	205.13.221.128	205.13.221.129	205.13.221.158	205.13.221.159
6	205.13.221.160	205.13.221.161	205.13.221.190	205.13.221.191
7	205.13.221.192	205.13.221.193	205.13.221.222	205.13.221.223
8	205.13.221.224	205.13.221.225	205.13.221.254	205.13.221.255

- Subnet 1 (205.13.221.0) is called the Subnet Zero.
- Subnet 8 (205.13.221.224) is called the All-Ones Subnet.
- We only need 6 LANs, so any 6 of these 8 subnets can be used — the other 2 remain spare for future growth.

Part 2: VLANs (Virtual LANs)

What is a VLAN?

A VLAN is a broadcast domain created by one or more switches. It logically groups ports/users regardless of their physical location, based on function, project team, or application — instead of grouping them by physical wiring.

- Traditionally, broadcast domains were separated using physical routers/segments (one LAN per switch/floor).
- VLANs let a single switch (or group of switches) host multiple separate broadcast domains.
- Rule: Switches never bridge traffic between VLANs — traffic between different VLANs must always go through a router (or Layer-3 device).
- Remember: VLAN = Subnet. A host must have an IP address from the correct subnet to properly belong to a VLAN.

Benefits of VLANs

- Logical (not physical) organization of the LAN.
- Easy to move, add, or reconfigure workstations without rewiring.
- Better control of broadcast traffic — smaller broadcast domains.
- Improved security, since VLANs isolate groups of users.

 **If a hub is connected to a switch port, every device on that hub must belong to the same VLAN as that port.**

VLAN Membership: Static vs Dynamic

Type	How it works
Static (port-based)	Admin manually assigns each switch port to a VLAN. Most common & simplest method. Works well with DHCP.
Dynamic	Switch queries a VLAN Management Database (e.g. CiscoWorks) for the MAC address of the connected device to decide VLAN membership. Rare today.

- Two-step process to place a host in the correct VLAN:
 - Connect the host to the correct switch port.
 - Assign the host the correct IP address for that VLAN's subnet.

VLAN Types

Type	Description
Port-based	Most common. Ports assigned individually/in groups to a VLAN. Simple; works well with DHCP.
MAC address-based	Rarely used today. Each MAC must be entered/configured manually — hard to manage.
Protocol-based	Similar to MAC-based but uses logical/IP address instead. No longer common because of DHCP.

Trunking and VLAN Tagging

- A trunk link does NOT belong to one specific VLAN — it carries traffic for multiple VLANs between switches/routers.
- VLAN tagging is needed whenever a single link must carry traffic for more than one VLAN.
- A tag/identifier is added to each frame header on the trunk to show which VLAN it belongs to; the tag is removed again before delivery to the end device.

Tagging Method	Notes
ISL (Inter-Switch Link)	Cisco-proprietary. Encapsulates the whole frame with an ISL header containing the VLAN ID — lengthens the frame. Older, less common now.
802.1Q	IEEE standard. Modifies (inserts into) the existing header instead of full encapsulation. This is the recommended/most common method today.

Native VLAN (802.1Q only)

- Exists for backward compatibility with old switches that don't understand VLAN tags.
- Frames belonging to the Native VLAN are sent untagged on trunk links.
- Any untagged frame received on a trunk port is automatically treated as belonging to the Native VLAN.
- By default, VLAN 1 is both the default VLAN and the Native VLAN — but this can be changed per port.

Configuring VLANs (Cisco Switch Commands)

- Guideline: max VLANs depend on the switch model (2900-series switches commonly allow up to 4095). VLAN 1 is always a factory-default VLAN (the default Ethernet VLAN).

Assign an access (non-trunk) port to a VLAN:

```
Switch(config)#interface fastethernet 0/9
```

```
Switch(config-if)#switchport access vlan 10
```

Create a VLAN via the VLAN database:

```
Switch#vlan database
```

```
Switch(vlan)#vlan 10
```

```
Switch(vlan)#exit
```

Force a port to stay an access port (not become a trunk):

```
Switch(config-if)#switchport mode access
```

 **Default port mode is 'dynamic desirable' — if connected to another switch with the same default mode, the link automatically becomes a TRUNK. Use 'switchport mode access' to prevent this on ports that should not trunk.**

Configure a range of ports at once (works on 2950, not all 2900 XL):

```
Switch(config)#interface range fastethernet 0/8, fastethernet 0/12
```

```
Switch(config-if)#switchport access vlan 3
```

Verifying VLANs

- show vlan — full details (name, status, ports, type, MTU, etc.)
- show vlan brief — quick summary of VLAN name, status, and ports

Deleting VLANs

Remove a port from a VLAN (port returns to default VLAN 1):

```
Switch(config-if)#no switchport access vlan 300
```

Delete an entire VLAN from the switch:

```
Switch#vlan database
```

```
Switch(vlan)#no vlan 300
```

```
Switch(vlan)#exit
```



VLAN 1 is the default/management VLAN and cannot be deleted.

Quick Recap

- Subnetting: borrow bits from host portion → more subnets, fewer hosts per subnet. Formula: usable hosts = $2^{(\text{host bits})} - 2$.
- VLAN = a logical broadcast domain = a subnet, independent of physical wiring.
- Switches never route between VLANs — only routers (or L3 switches) do.
- Trunk links carry multiple VLANs using tagging (ISL or 802.1Q — 802.1Q is preferred).
- Native VLAN = untagged traffic on a trunk (default = VLAN 1).